EASY · ADVANCED MATH

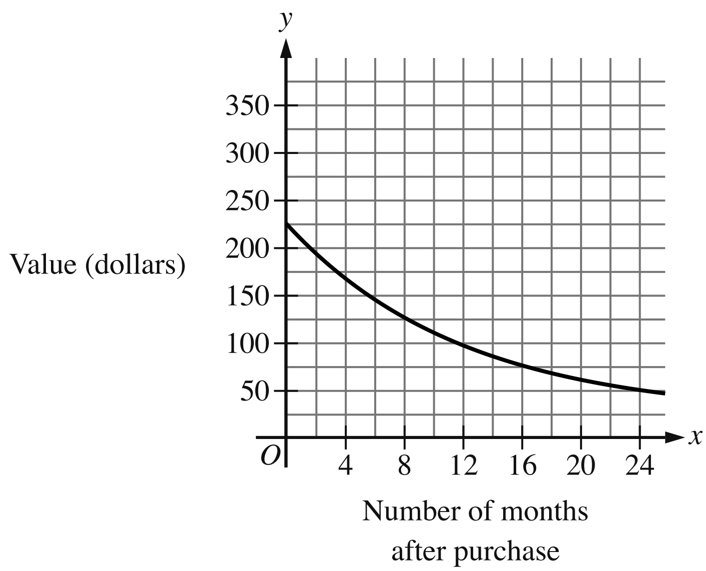
Advanced Math

14

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29



- Moving from left to right:
  - The curve is in quadrant 1.
  - The curve trends down gradually.
- The curve passes through the following approximate points:
  - (0 comma 225)
  - (24 comma 50)

The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

A. The estimated value of the tablet was $\$225$ when it was purchased.

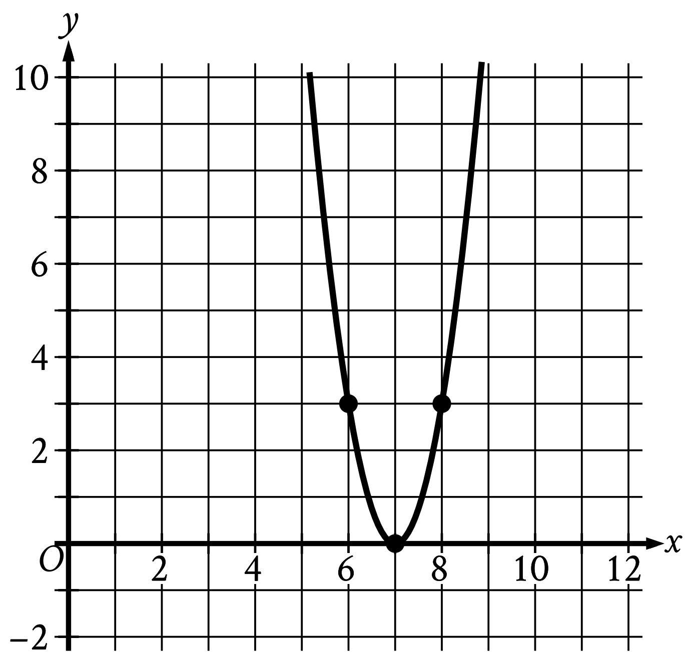
B. The estimated value of the tablet 24 months after it was purchased was $\$225$.

C. The estimated value of the tablet had decreased by $\$225$ in the 24 months after it was purchased.

D. The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

Q2

GRID-IN ADVANCED MATH



- The parabola opens upward.
- The vertex is at point (7 comma 0).
- The parabola passes through the following points:
  - (6 comma 3)
  - (7 comma 0)
  - (8 comma 3)

The x-intercept of the graph shown is $x, 0$. What is the value of x?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ADVANCED MATH

The function f is defined by $f x = 8\sqrt{x}$. For what value of x does $f x = 48$?

A. 6

B. 8

C. 36

D. 64

Q4

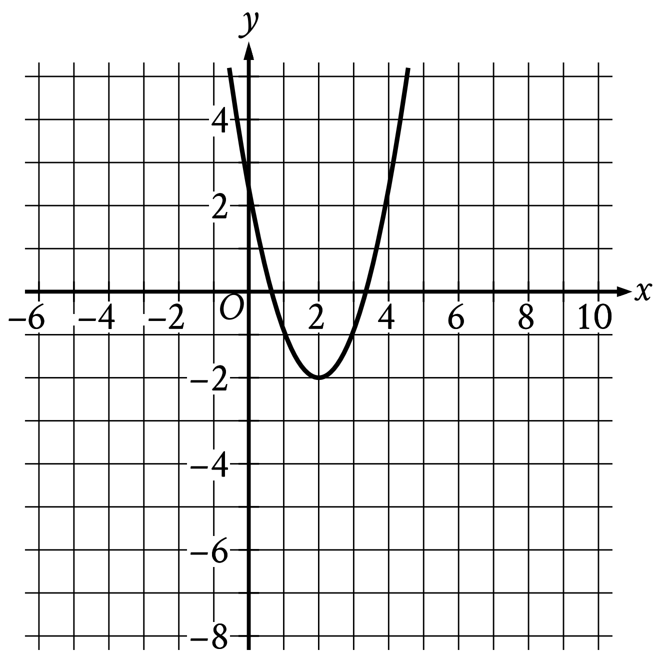
GRID-IN ADVANCED MATH

The function f is defined by $f x = x^2 + x + 71$. What is the value of $f 2$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

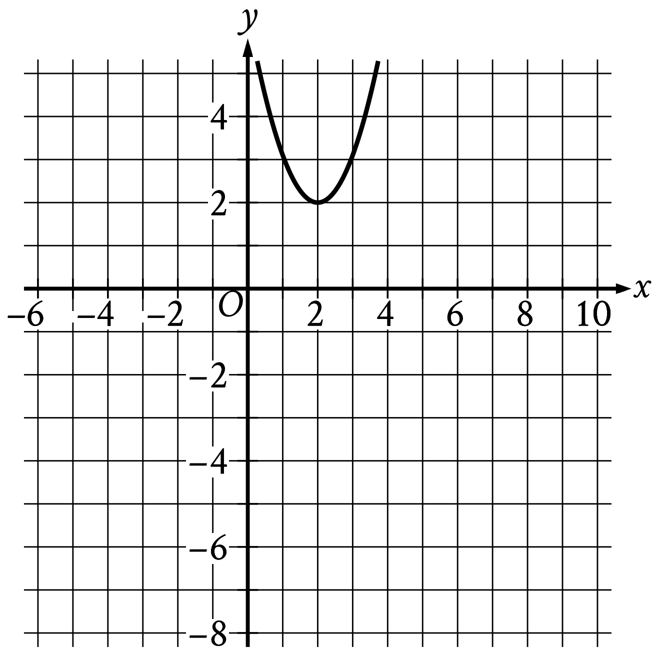
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The parabola opens upward.
- The vertex is at the point (2 comma negative 2).
- The parabola passes through the following points:
    - (1 comma negative nine tenths)
    - (2 comma negative 2)
    - (3 comma negative nine tenths)

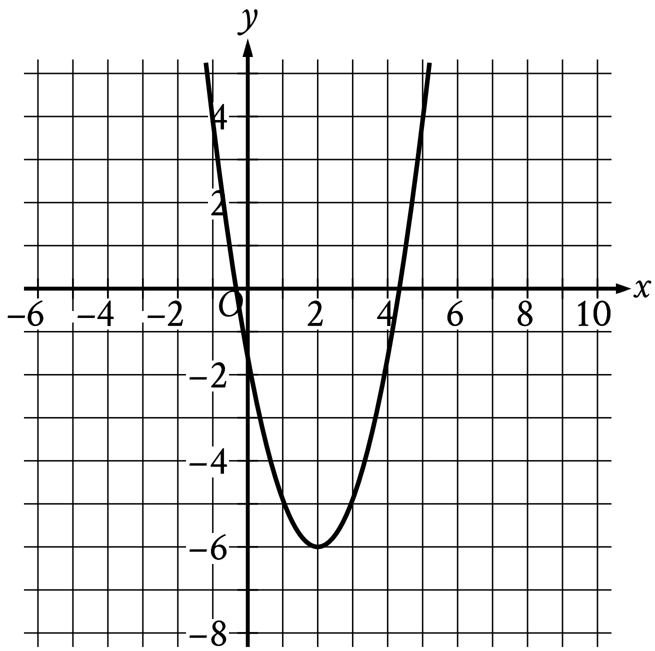
The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

A.



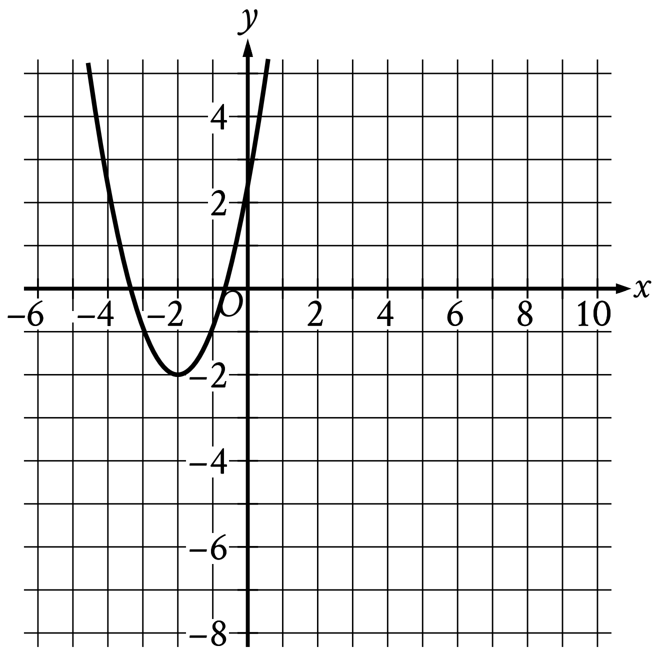
- The parabola opens upward.
- The vertex is at the point (2 comma 2).
- The parabola passes through the following points:
  - (1 comma StartFraction 31 Over 10 EndFraction)
  - (2 comma 2)
  - (3 comma StartFraction 31 Over 10 EndFraction)

B.



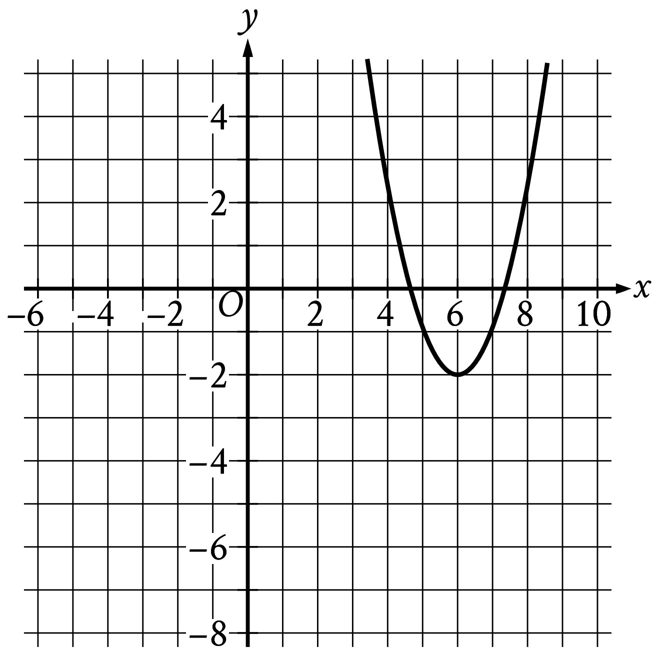
- The parabola opens upward.
- The vertex is at the point (2 comma negative 6).
- The parabola passes through the following points:
  - (1 comma negative StartFraction 49 Over 10 EndFraction)
  - (2 comma negative 6)
  - (3 comma negative StartFraction 49 Over 10 EndFraction)

C.



- The parabola opens upward.
- The vertex is at the point (negative 2 comma negative 2).
- The parabola passes through the following points:
  - (negative 3 comma negative nine tenths)
  - (negative 2 comma negative 2)
  - (negative 1 comma negative nine tenths)

D.



- The parabola opens upward.
- The vertex is at the point (6 comma negative 2).
- The parabola passes through the following points:
  - (5 comma negative nine tenths)
  - (6 comma negative 2)
  - (7 comma negative nine tenths)

Q6 ADVANCED MATH

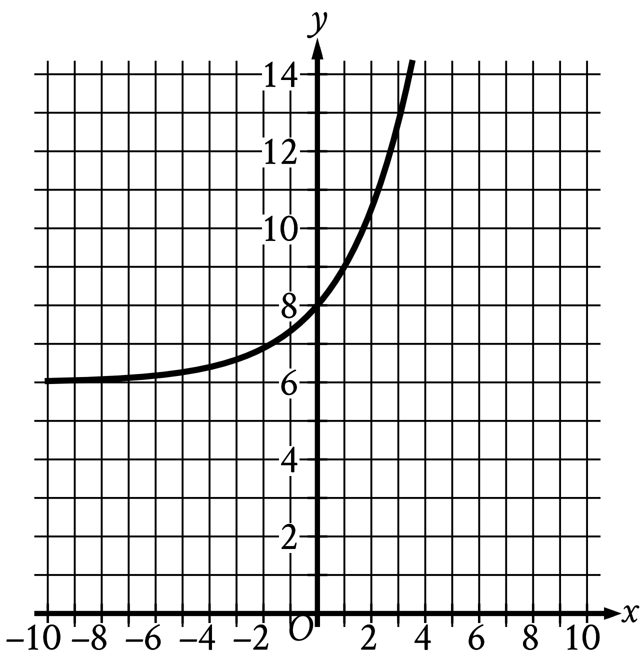
The kinetic energy, in joules, of an object with mass 9 kilograms traveling at a speed of v meters per second is given by the function K, where $Kv = \frac{9}{2}v^2$. Which of the following is the best interpretation of $K34 = 5{,}202$ in this context?

A. The object traveling at 34 meters per second has a kinetic energy of 5,202 joules.

B. The object traveling at 340 meters per second has a kinetic energy of 5,202 joules.

C. The object traveling at 5,202 meters per second has a kinetic energy of 34 joules.

D. The object traveling at 23,409 meters per second has a kinetic energy of 34 joules.



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to point (0 comma 8).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - (0 comma 8)
  - (1 comma 9)

What is the y-intercept of the graph shown?

A. $-8, 0$

B. $-6, 0$

C. $0, 6$

D. $0, 8$

Q8

GRID-IN ADVANCED MATH

The y-intercept of the graph of $y = x^2 + 31$ in the xy-plane is $0, y$. What is the value of y?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 ADVANCED MATH

The function f is defined by $f x = x^3 + 15$. What is the value of $f 2$?

A. 20

B. 21

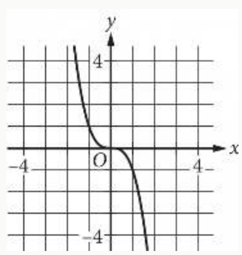
C. 23

D. 24

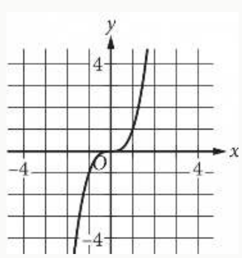
x	y
0	0
1	1
2	8
3	27

The table shown includes some values of x and their corresponding values of y. Which of the following graphs in the xy-plane could represent the relationship between x and y ?

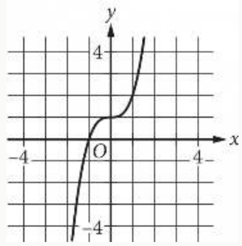
A.



B.



C.



D.